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Grant M. Plummer; Eric Herbst; Frank C. De Lucia

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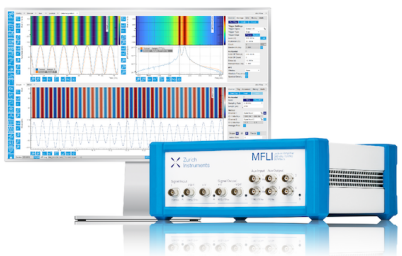
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Submillimeter spectra and molecular constants of ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD[a)]

Grant M. Plummer, Eric Herbst, and Frank C. De Lucia
Department of Physics, Duke University, Durham, North Carolina, 27706

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The fundamental $J = 1 \leftarrow 0$ rotational transitions of ^{7}LiH in its two lowest vibrational states and of ^{6}LiH in its ground vibrational state have been measured. In addition, the $J = 2 \leftarrow 1$ transitions of ^{7}LiD in its two lowest vibrational states and of ^{6}LiD in its ground vibrational state have been studied. The measured submillimeter transition frequencies have been combined with previous millimeter data on ^{6}LiD and ^{7}LiD into a data set which has been successfully analyzed according to the theories of Herman, Asgharian, and Watson, which treat the problem of Born–Oppenheimer breakdown.

I. INTRODUCTION

Despite the attractive simplicity of alkali hydride systems, the pure rotational spectra of only four such species—^{6}LiD, ^{7}LiD, NaH, and NaD—have previously been observed.[1,2] The basic difficulty, particularly with the heavier species, lies in the production of detectable concentrations of the hydride vapor from solid samples at the low pressures necessary to maintain microwave resolution. Microwave[3] and glow[2] discharges have been employed to produce samples of NaH and NaD for optical and millimeter/submillimeter studies. Fortunately, the conditions are somewhat more favorable in the case of the lithium hydrides. Pearson and Gordy[1] measured four millimeter transitions of ^{6}LiD and ^{7}LiD by simply heating the solid, though at the time of that experiment the sensitivity necessary at the considerably higher LiH frequencies was not attainable. Recent years have seen increases in the sensitivity of microwave techniques, both in the laboratory[4,5] and in the receivers used for astronomical observations.[6,7] In light of these advances and the possibility that LiH may exist in certain carbon-rich red giant stars,[8,9] we have constructed a hot cell similar to that used by Pearson and Gordy and successfully measured three $J = 1 \leftarrow 0$ transitions: those of ^{7}LiH in its ground and first excited vibrational states and that of ^{6}LiH in its ground vibrational state. We have also measured the $J = 2 \leftarrow 1$ transitions of ^{7}LiD in its ground and first excited vibrational states and of ^{6}LiD in its ground state.

These species are of particular theoretical interest by virtue of their small masses. A breakdown in the standard Dunham treatment[10] of diatomic rotation-vibration spectra is easily observable in light molecules, but can be treated successfully with several higher order corrections[11,12] if sufficient isotopic information is available. Our observations have allowed us to accurately determine these corrections for LiH/LiD and to reproduce the observed spectra to within the experimental uncertainty (50 KHz).

II. EXPERIMENTAL

We have discussed elsewhere[2] some of the difficulties of performing low pressure spectroscopy on alkali hydrides. For this experiment a nickel cell similar to that used for earlier work on LiD[1] was constructed. The cell, 5 ft in length with a 2 in. inner diameter, was fitted with Teflon end plates, which are transparent throughout the millimeter and submillimeter. Approximately 5 g of either LiH or LiD in powdered form was placed in the center of the nickel jacket. Ceramic heating elements were used to slowly raise this center section to ~ 1000 K as measured by an external chromel–alumel probe, while the ends of the cell were cooled to protect the windows and vacuum seals. Microwave radiation produced by a klystron-driven crossed guide harmonic generator was focused through the cell quasi-optically and detected by a 1.5 K InSb detector. A more complete description of the spectrometer and multiplier can be found in Refs. 4 and 5.

As the indicated cell temperature approached 1000 K, decomposition of the hydride into Li(1) and H_2(g) produced a background pressure of several Torr. To avoid excessive broadening of the absorption lines, the cell was continuously pumped to maintain a total pressure of less than 500 mTorr, of which the desired hydride vapor was only a small fraction (10^{-3} or 10^{-4}).[2] These hydrides, however, possess rather large dipole moments,[13] and despite the small concentration of ^{7}LiH(g), its $J = 1 \leftarrow 0$ ground vibrational state line was detected during a real-time video search with a S/N ratio of approximately two. All other transitions measured during the course of this work were weaker by one or two orders of magnitude, due to the smaller natural abundance of ^{6}Li (7.4%), lower excited state population, or both. To record these lines the klystron was phase locked, frequency modulated, and swept through the transitions. Signals were detected at $2f$ with a lock-in amplifier ($\tau = 30$ ms) and digitally integrated for aproximately 10 min. In general, lines were fairly strong for a period of 30 to 45 min per run, during which time the production of H_2 reached a maximum and then dropped steadily. Because the H_2 pressure was lower than expected at the measured external temperature of 1000 K, it is probable that the actual sample temperature was somewhat lower. This suspicion was supported by an examination of the sample chamber, performed after several runs had been completed, which showed that only small portions of the sample had actually melted or decomposed into Li(1) during the runs.

III. THEORETICAL BACKGROUND

The standard treatment of the rotational-vibrational levels of $^1\Sigma$ diatomics calls for application of the Dunham expansion[10]

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$$E(v,J) = h\sum_{j,k} Y_{jk}(v + 1/2)^j J^k (J+1)^k. \quad (1)$$

Transition frequencies within the vibrational state v for $J+1 \leftarrow J$ are then given by

$$\nu(v;J+1\leftarrow J) = 2Y_{01}(J+1) + 2Y_{11}(v+1/2)(J+1) + 2Y_{21}(v+1/2)^2(J+1) + 4Y_{02}(J+1)^3 + 4Y_{12}(v+1/2)(J+1)^3 + \cdots. \quad (2)$$

Within the Born–Oppenheimer approximation the Y_{jk} are related to their counterparts in isotopically substituted species according to the ratios of their reduced masses raised to appropriate powers:

$$Y'_{jk}/Y_{jk} = [\mu/\mu']^{(j+2k)/2}. \quad (3)$$

This is not an exact expression within the Dunham theory, but is accurate enough for our purposes except in application to Y_{01}. Though the prediction of Eq. (3) for a μ^{-1} dependence does not hold for Y_{01}, Dunham has expressed Y_{01} in terms of the Born–Oppenheimer equilibrium constant B_e^{BO} and an additive correction term ΔY_{01}:

$$Y_{01} = B_e^{BO} + \Delta Y_{01}. \quad (4)$$

The mass dependencies of these quantities are

$$\Delta Y_{01} \propto \mu^{-2}, \quad (5)$$

$$B_e^{BO} \propto \mu^{-1}, \quad (6)$$

the former because $\Delta Y_{01} \propto (B_e^{BO})^3/Y_{10}^2$, and the latter by virtue of the fact that the Born–Oppenheimer equilibrium bond length (r_e^{BO}), which defines B_e^{BO} through

$$B_e^{BO} = h/8\pi^2\mu(r_e^{BO})^2, \quad (7)$$

is constant for the different isotopic structures.

Such relations between the rotational constants of isotopically substituted species are of great importance in the analysis of light molecules. Because their rotational transitions fall at high frequencies, measurement of the four or five transitions which are necessary to determine the larger rotational constants is often impossible. However, some transition frequencies of the related species are usually accessible and, if an accurate relationship between the constants of the related species is known, can provide the information necessary to determine the complete set of constants. Unfortunately, since the Born–Oppenheimer approximation is essentially an expansion in powers of $(m_{\text{electronic}}/M_{\text{nuclear}})^{1/4}$, the accuracy of the model outlined above when applied to isotopically substituted species of diatomic molecules decreases as the atomic masses decrease. Careful analyses[14,15] show that, even in the case of heavy diatomics, relations (2)–(6) do not hold to within the accuracy of microwave spectroscopic techniques. Large errors arise in the analyses of light species, and use of a more sophisticated model of the relationship between rotational constants of the different species is necessary if the isotopic information is to be exploited.

In a treatment of the Born–Oppenheimer breakdown observed in these lighter diatomics, Herman and Asgharian[11] have shown that to first order adiabatic effects introduce two correction terms to the Y_{jk} which depend separately on the inverses of the atomic masses. The following arguments are due to Watson,[12] who has simplified the rotational analysis by rewriting the nonadiabatic corrections to Y_{01} in similar form. Watson has shown that the equilibrium bond lengths r_e, which are given by

$$r_e = (h/8\pi^2\mu Y_{01})^{1/2}, \quad (8)$$

and which differ from species to species, are related to the isotopically invariant Born–Oppenheimer bond lengths r_e^{BO} according to

$$r_e = r_e^{BO}\left[1 + \frac{m_e d_a}{M_a} + \frac{m_e d_b}{M_b}\right], \quad (9)$$

where m_e is the electronic mass, and M_a and M_b are the atomic masses. The isotopically invariant quantities d_a and d_b consist of three terms each:

$$d_a = d_a^{ad} - \frac{\mu\,\Delta Y_{01}}{2m_e B_e^{BO}} - \frac{(\mu g_j)_b}{2m_p}$$

$$d_b = d_b^{ad} - \frac{\mu\,\Delta Y_{01}}{2m_e B_e^{BO}} - \frac{(\mu g_j)_a}{2m_p}. \quad (10)$$

Here m_p is the proton mass and ΔY_{01} and B_e^{BO} are the Born–Oppenheimer quantities mentioned above. According to Eqs. (5)–(6), the quantity $\{\mu Y_{01}/B_e^{BO}\}$ is constant for the different isotopic structures.

The quantities $(\mu g_j)_a$ and $(\mu g_j)_b$ are invariant only within the "point dipole" approximation[16,17] which allows the calculation of the g_j factors of the substituted species from the g_j value and the electric dipole moment of a parent species. The symbols $(\mu g_j)_i$ are the values of the μg_j calculated in coordinate systems fixed at the nuclei. The center-of-mass g_j factor is given by

$$g_j = (\mu g_j)_b/M_a + (\mu g_j)_a/M_b. \quad (11)$$

The remaining terms—d_a^{ad} and d_b^{ad}—are the true adiabatic corrections to the bond length. They are given by Watson[12] in their simplest form in terms of the Born–Oppenheimer force constant f as

$$d_i^{ad} = -(2m_e f r_e^{BO})^{-1}\left[d/dr\langle n|P_i^2|n\rangle\right]_{r_e^{BO}}, \quad (12)$$

where $i = a,b$. The matrix elements of the nuclear momenta P_i referenced to the space-fixed frame between the electronic states $|n\rangle$ are independent of the isotopic species under consideration.

For each species the spectroscopically determined Y_{01} can be derived from expressions (8), (9), and (10) to be

$$Y_{01} = B_e^{BO} + \frac{\Delta_{01}^a}{M_a} + \frac{\Delta_{01}^b}{M_b} \quad (13)$$

with

$$\Delta_{01}^a = -2m_e d_a B_e^{BO} = -2m_e d_a^{ad} B_e^{BO} + \mu\Delta Y_{01} + \frac{m_e}{m_p}(\mu g_j)_b B_e^{BO}, \quad (14a)$$

$$\Delta_{01}^b = -2m_e d_b B_e^{BO} = -2m_e d_b^{ad} B_e^{BO} + \mu\Delta Y_{01} + \frac{m_e}{m_p}(\mu g_j)_a B_e^{BO}. \quad (14b)$$

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TABLE I. Measured, calculated, and predicted frequencies[a] (MHz).

Species	$J' \leftarrow J$	v	Observed	$(o-c)^b$	$(o-p)^c$	$(o-p)^d$	$(o-p)^e$
^{6}LiH	1←0	0	453 160.28	0.03	0.09	− 37.8	± 7[f]
^{7}LiH	1←0	0	443 952.80	− 0.03	− 0.10	− 37.1	− 3.69
	1←0	1	431 217.57	0.00	3.87	− 30.9	+ 3.69
^{6}LiD	1←0	0	260 306.96	0.01	0.02	0.0	0.04
	1←0	1	254 596.63	− 0.05	− 0.10	0.0	− 0.09
	2←1	0	520 404.55	− 0.01	− 0.03	0.0	0.07
^{7}LiD	1←0	0	251 403.53	− 0.01	− 0.05	0.0	− 0.04
	1←0	1	245 635.62	0.05	0.10	0.0	0.09
	2←1	0	501 892.43	0.01	0.03	0.0	− 0.06
	2←1	1	491 079.07	0.00	− 1.34	0.1	− 7.39

[a] Frequencies below 430 GHz from Ref. 1.
[b] Based on constants in Table II.
[c] Based on fits excluding the predicted transition. See the text.
[d] Based on constants of Vidal and Swalley, Ref. 18.
[e] Based on wobble-stretch fit excluding the predicted transition.
[f] Estimated error due to point dipole approximation. See the text.

We note that the additive corrections Δ_{01}^{i} exhibit the mass dependence μ^{-1}.

IV. ANALYSIS AND RESULTS

Vidal and Stwalley[18] have completed an analysis of the previously available optical, infrared, and microwave data pertaining to the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of LiH and LiD according to the approach presented by Herman and Asgharian.[11] Their conclusion is that the effects of the Born–Oppenheimer breakdown on the ground electronic state pure rotational spectrum can be accounted for by the above corrections to Y_{01} alone, though the larger effects of the same sort witnessed in the excited electronic state spectrum require that similar corrections be applied to the higher order rotational constants as well. Similarly, we have analyzed our microwave data by applying Eq. (3) to the higher order Dunham constants and using additive corrections to Y_{01} via Eq. (13).

The six submillimeter lines measured in the course of this work are listed in Table I along with the four $J = 1 \leftarrow 0$ lines of ^{6}LiD and ^{7}LiD measured by Pearson and Gordy.[1] A least squares analysis including all these observations was carried out utilizing the modified Dunham expression

$$\nu'(v;J+1\leftarrow J) = 2R'(J+1)\left[B_e^{BO} + \frac{\Delta_{01}^{H}}{M'_H} + \frac{\Delta_{01}^{Li}}{M'_{Li}}\right]$$
$$+ 2R'^{3/2}(v+1/2)(J+1)Y_{11} + 2R'^{2}(v+1/2)^2(J+1)\,Y_{21}$$
$$+ 4R'^{2}(J+1)^3Y_{02} + 4R'^{5/2}(v+1/2)(J+1)^3Y_{12}, \qquad (15)$$

where the primes denote quantities which are isotopically dependent, $R' = \mu(^6\text{LiH})/\mu'$, and the unprimed constants are those of the species ^{6}LiH. The values of the constants determined in the fit are shown in Table II, along with uncertainties which correspond to one standard deviation. The residuals listed in Table I combine to yield an rms deviation of 46 KHz, which is less than the estimated measurement uncertainty of 50 KHz.

In addition, sets of nine out of the ten data points were analyzed and the resulting constants used to "predict" the frequency of the tenth ("pulled") line. The results, listed in Table I, add to our confidence in the assignments and analysis. Only two of the ten predictions fall farther than 100 KHz from the measured values, and in those two cases the uncertainties in the constants increase so that the measured frequencies lie within the uncertainty of the predictions.

The last columns of Table I list the differences between the observed frequencies and the frequencies calculated from the ground state constants of Vidal and Stwalley,[18] and from the constants derived from a pull-predict fit using the wobble-stretch approach, which is discussed in detail in Sec. V. In light of the fact that no microwave data on the hydride species was available to Vidal and Stwalley at the time of their analysis, the errors listed are reasonably small. However, our analysis of the ten microwave transitions alone leads to substantial changes in the values of the constants B_e^{BO}, Y_{11}, Y_{21}, and Δ_{01}^{H}, which may be due in part to the necessary omission of some higher order constants in our

TABLE II. Molecular constants.

	^{6}LiH	^{7}LiH	^{6}LiD	^{7}LiD
B_e^{BO}(MHz)	230 161.654(157)	225 449.805(153)	131 671.220(90)	126 959.371(87)
Δ_{01}^{H}(MHz amu)	− 195.677(44)	− 191.671(43)	− 111.943(25)	− 107.937(24)
Δ_{01}^{Li}(MHz amu)	− 14.596(816)	− 14.297(800)	− 8.350(467)	− 8.051(450)
Y_{11}(MHz)	− 6 695.380(224)	− 6 490.835(219)	− 2 897.090(125)	− 2 742.982(121)
Y_{21}(MHz)	63.733(107)	61.151(104)	20.859(61)	19.392(59)
Y_{02}(MHz)	− 26.832(16)	− 25.745(16)	− 8.782(9)	− 8.164(9)
Y_{12}(MHz)	0.476(20)	0.452(20)	0.118(12)	0.108(11)
Y_{01}[a](MHz)	229 965.070(210)	225 257.584(172)	131 614.252(120)	126 904.633(108)

[a] Calculated using Eq. (13).

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analysis. A comparison of the larger rotational constants for ^{7}LiH is listed in Table III.

V. DISCUSSION AND CONCLUSION

By combining terms among relations (13) and (14) and using Eq. (11) and the previously discussed isotopic mass relations, it can be seen that Y'_{01} is given as

$$Y'_{01} = R'B_e^{BO} - R'B_e^{BO}\left[\frac{2m_e d_a^{ad}}{M'_a} + \frac{2m_e d_b^{ad}}{M'_b}\right] + R'^2\Delta Y_{01} + R'\frac{m_e}{m_p}g'_j B_e^{BO}. \quad (16)$$

The unprimed rotational constants belong to a particular isotopic species (^{6}LiH), and primes denote the quantities which vary from species to species. This form lends itself to an algebraic comparison of the Herman–Asgharian approach and the wobble-stretch method of adiabatically correcting Y_{01}, which was introduced by Rosenblum, Nethercot, and Townes[19] in their analysis of the Born–Oppenheimer breakdown in CO. According to this approach the correction term, a second order perturbation correction for the electronic-rotational interaction, exhibits the mass dependence μ^{-2}. If we label the corrections applicable to each species (ws)', the wobble-stretch expression for Y'_{01} is

$$Y'_{01} = R'B_e^{BO} + R'^2(\mathrm{ws}) + R'^2\Delta Y_{01} + R'\frac{m_e}{m_p}g'_j B_e^{BO}, \quad (17)$$

where

$$(\mathrm{ws})' = \mathrm{R}'^2(\mathrm{ws}) = R'\left[\frac{\mu(\mathrm{ws})}{M'_a} + \frac{\mu(\mathrm{ws})}{M'_b}\right]. \quad (18)$$

Comparison of expressions (16) and (17)–(18) illustrates the fact that the two approaches are equivalent only when $d_a^{ad} = d_b^{ad}$, as in the case of homonuclear diatomics.

That the atomic mass scaling approach is indeed more accurate than the wobble-stretch correction has been demonstrated directly by other workers in their analysis of CO,[12] PbS,[14] GeS, and GeSe.[15] Though in these heavier species the effects are much smaller, the large numbers of accessible transitions at millimeter frequencies allow determination of Y_{01} for individual isotopic species without the use of isotopic relations. In such cases a direct and unambiguous comparison of the two approaches can be conducted. The parameters d_a^{ad} and d_b^{ad} can be obtained using Eq. (10) from measured values of the dipole moment and g_j factor of one isotope, which in the point-dipole approximation yield all the $(\mu g_j)_i$, and from the Born–Oppenheimer expressions for ΔY_{01}[17]:

$$\begin{aligned}\Delta Y_{01} &= (B_e^{BO}/Y_{10}^2)\beta_{01},\\ \beta_{01} &= Y_{01}^2(Y_{21}/4Y_{01}^3) + 16a_1(Y_{20}/3Y_{01}) - 8a_1 - 6a_1^2 + 4a_1^3, \quad (19)\\ a_1 &= Y_{11}Y_{10}/6Y_{01}^2 - 1.\end{aligned}$$

A statistically meaningful difference between d_a^{ad} and d_b^{ad}, such as that found in CO, PbS, GeS, and GeSe, serves as direct evidence in support of the Herman–Asgharian approach. With analyses such as ours, subject to a paucity of observable transitions which prohibits determination of Y_{01} and the higher order constants in the absence of an expression relating the Y_{jk} in the different species, such direct comparison is not possible.

Other analyses of the molecules LiH and CO, however, strongly support the theory of Herman and Asgharian[11] by showing that it allows acceptable fits to their observed spectra in cases where corrections based on reduced mass ratios do not. Laser frequency measurements[20] of optical transitions have shown that the corrections to the constant Y_{10} of CO must consist of terms dependent on the atomic masses alone. The work of Vidal and Stwalley[18] also offers strong evidence of the superiority of this approach in that the rms deviation of their fit to a large data set including microwave and optical transitions is much smaller than that obtained with the same LiH data set using a correction scheme based on reduced mass ratios.

We have found that, albeit within the constraints of our smaller data set, the same is true for the pure rotational spectrum. Analyses of our data according to Eq. (17) have been carried out in which the constants $\{(\mathrm{ws}) + \Delta Y_{01}\}$ and B_e^{BO} were varied while the g_j factors for all isotopes were held at values calculated using Eq. (11) from measured values of the electric dipole moment and g_j factor of a particular species.[13,21] These parameters have been measured for the ^{7}Li species only, and different values for the g_j are obtained from

TABLE III. Comparisons with previous values of ^{7}LiH constants.

	This work	Vidal and Stwalley[a]	Pearson and Gordy[b]
B_e^{BO} (MHz)	225 449.805(153)	225 428.85(2.34)	225 452(7)
Δ_{01}^{H} (MHz amu)	− 191.671(43)	− 155.3(4.5)	···
Δ_{01}^{Li} (MHz amu)	− 14.297(800)	− 14.34(48)	···
Y_{01} (MHz)	225 257.584(172)	225 272.7(5.0)	225 256(5.5)
Y_{11} (MHz)	− 6 490.835(219)	− 6 480.17(42)	− 6 495(6)
Y_{21} (MHz)	61.151(104)	53.252(330)	···
Y_{31} (MHz)	···	0.870(51)	···
Y_{41} (MHz)	···	− 0.161(2)	···
Y_{02} (MHz)	− 25.745(16)	− 25.777(9)	···
Y_{12} (MHz)	0.452(20)	0.477(2)	···

[a] Obtained from Ref. 18.
[b] From Ref. 1.

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the two sets of values. The values of g_j obtained from the available ground vibrational state ^{7}LiH data are $-0.640\ 10$ and $-0.256\ 15$ for ^{6}LiH and ^{6}LiD, respectively, and use of the ground vibrational state ^{7}LiD data leads to the values $-0.643\ 88$ and $-0.258\ 57$. The rms deviations of the fits using either set of results for the g_j values were an order of magnitude higher than those obtained with Eq. (15). Additional fits using Eq. (17) in which only the ^{7}LiH and ^{7}LiD g_j factors were fixed at measured values show that the g_j values of *both* ^{6}Li species must be varied if the rms is to be brought down to acceptable levels. Since in these cases four independent variables—$\{(\mathrm{ws}) + \Delta Y_{01}\}$, B_e^{BO}, and two g_j factors—are used to calculate the four rotational constants Y'_{01}, *each isotopic structure has, in essence, its own independent* Y'_{01}. In constrast, the Watson approach does as well using only three independent constants to determine the four Y'_{01} values.

Also, the approach using Eq. (17) was less successful than the Watson approach in the pull-predict analysis described in Sec. IV, as is shown in Table I. Note that it is not possible to predict the ^{6}LiH frequency on the basis of the other nine data points since in this analysis the independent variable $g_j(^6\mathrm{LiH})$ is determined entirely by that transition. The value listed in Table I for this transition is an estimate of the error in $2Y_{01}$ due to of an error of 1% in the point dipole value for $g_j(^6\mathrm{LiH})$. This estimate is based on the discrepancy of approximately 1%, noted above, between calculated values of the g_j factors using data of different species, which serves as a rough limit on the accuracy of the point dipole model.

We note that the approximate nature of the point dipole model leaves our arguments against the wobble-stretch approach somewhat inconclusive. When the ^{6}Li species' g_j factors are allowed to vary in a fitting procedure using the wobble-stretch correction and incorporating measured ground vibrational state values of the ^{7}Li species g_j factors, the values obtained are $-0.642\ 09(25)$ and $-0.260\ 81(17)$. The parameters are well determined and lie within the estimated error of the point dipole model, even though ground vibrational state dipole moments and molecular g_j factors are assumed here to apply to the excited vibrational state as well. Thus, despite the apparent advantages of the atomic mass scaling approach, the present analysis does not conclusively prove the wobble-stretch approach to be in error. Watson's formulation of the Herman–Asgharian theory also incorporates the point dipole approximation, but the g_j factors are not directly obtained in the analysis. Since they appear only in linear combinations with ΔY_{01} and the d_i^{ad}, the effect of the point dipole error is contained in the uncertainties of those linear combinations and is not directly observable. It is clear, however, that this error must be considered in any interpretation of the constants Δ_{01}^{H} and $\Delta_{01}^{\mathrm{Li}}$.

In conclusion, our study shows that the size of the Born–Oppenheimer breakdown in the LiH/LiD system is, as expected, larger than in any of the other diatomics which have been so analyzed.[22,23] There is evidence that the theory of Herman and Asgharian[11] as extended by Watson[12] is correct since two independent terms which vary as the inverse of the atomic masses are adequate to correct the Born–Oppenheimer breakdown for this light diatomic, and since the theory provides reliable predictions of measured transition frequencies when they are excluded from the data set. The correction scheme based on reduced mass scaling is less satisfactory in that it requires an additional degree of freedom to accurately reproduce the observed rotational spectrum of LiH and seems a less accurate predictive tool.

[1] E. F. Pearson and W. Gordy, Phys. Rev. **177**, 59 (1969).
[2] K. V. L. N. Sastry, E. Herbst, and F. C. De Lucia, J. Chem. Phys. **75**, 4753 (1981).
[3] M. Giroud and O. Nedelec, J. Chem. Phys. **73**, 4151 (1980).
[4] F. C. De Lucia, in *Molecular Spectroscopy, Modern Research*, edited by K. N. Rao (Academic, New York, 1976), Vol. II.
[5] P. Helminger, J. K. Messer, and F. C. De Lucia, Appl. Phys. Lett. **42**, 309 (1983).
[6] T. G. Phillips and K. B. Jefferts, IEEE MTT **22**, 1290 (1974).
[7] E. C. Sutton, IEEE MTT **31**, 589 (1983).
[8] H. Reeves, Annu. Rev. Astron. Astrophys. **12**, 437 (1974), and references therein.
[9] G. M. Plummer, E. Herbst, and F. C. De Lucia, Astrophys. J. Lett. **282**, L113 (1984).
[10] J. L. Dunham, Phys. Rev. **41**, 721 (1932).
[11] R. M. Herman and A. Asgharian, J. Mol. Spectrosc. **19**, 305 (1966).
[12] J. K. G. Watson, J. Mol. Spectrosc. **45**, 99 (1973).
[13] L. Wharton, L. P. Gold, and W. Klemperer, J. Chem. Phys. **33**, 1255 (1960); E. Rothstein, *ibid.* **50**, 1899 (1969); **52**, 2804 (E) (1970).
[14] E. Tiemann, W. U. Stieda, T. Torring, and J. Hoeft, Z. Naturforsch. Teil A **30**, 1606 (1975).
[15] W. U. Steida, E. Tiemann, T. Torring, and J. Hoeft, Z. Naturforsch. Teil A **31**, 374 (1976).
[16] W. Gordy and R. L. Cook, *Microwave Molecular Spectroscopy* (Wiley–Interscience, New York, 1970), p. 401.
[17] C. H. Townes and A. L. Schawlow, *Microwave Spectroscopy* (McGraw–Hill, New York, 1955), pp. 295 and 296.
[18] C. R. Vidal and W. C. Stwalley, J. Chem. Phys. **77**, 883 (1982).
[19] B. Rosenblum, A. H. Nethercot, and C. H. Townes, Phys. Rev. **109**, 400 (1957).
[20] A. H. M. Ross, R. S. Eng, and H. Kildal, Opt. Commun. **12**, 433 (1974).
[21] R. R. Freeman, A. R. Jacobson, D. W. Johnson, and N. F. Ramsey, J. Chem. Phys. **63**, 2597 (1975).
[22] E. F. Pearson and W. Gordy, Phys. Rev. **177**, 52 (1969).
[23] F. C. De Lucia, P. Helminger, and W. Gordy, Phys. Rev. A **3**, 1849 (1971); See also Refs. 1, 2, 11, 14, and 16–20.

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